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About this Course

Key Components of Llamafile

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Reading: Key Terms
5 min
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Reading: What is Llamafile?
5 min
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Video: Llamafile overview by Mozilla
5 min
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Video: Using the Llamafile API
2 min
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Video: Creating a Llamafile
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Reading: Cosmopolitan
5 min
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Video: Building portable binaries with Cosmopolitan
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Lab: Cosmopolitan
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Reading: Lesson Reflection
5 min
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Graded Assignment: Quiz- Key Components of Llamafile
10 min

Developing Portable CLI with Cosmopolitan

Running Llamafile

Conclusion and Next Steps

Lesson Reflection

Lesson Summary

This lesson introduced the Cosmopolitan Libc, which enables writing C programs that can be compiled once and run anywhere across platforms like Linux, Windows, and MacOS. Key concepts included using the custom cosmopolitan compiler toolchain, debugging facilities like strace and ftrace, building portable binaries from simple hello world programs to command line tools, and understanding how Cosmopolitan integrates with distribution tools like LlamaFile for sharing large language models.

Key Points

- Cosmopolitan provides Java-like portability for C code
- Special cosmo compiler outputs portable binaries
- Print system traces and enable logging for debugging
- Build cross-platform C programs like Marco Polo CLI
- Core part of LlamaFile for distributing models

Reflection Questions

1. How does Cosmopolitan provide "write once, run anywhere" portability for C programs?
2. What is the benefit of using the custom Cosmopolitan toolchain over normal gcc/clang?
3. What debugging and tracing facilities does Cosmopolitan offer? When might they be useful?
4. How would you structure a cross-platform command line tool in C using Cosmopolitan?
5. Why is Cosmopolitan integration important for distributing large language models with LlamaFile?

Challenge Exercises

1. Compile and run a simple Cosmopolitan "Hello World" on multiple OSes
2. Build a command line tool with argument parsing and help text
3. Enable ftrace debug logging and analyze the output
4. Run a Linux sysadmin tool like ps or ls using Cosmopolitan
5. Modify an existing C utility/program to use the Cosmopolitan toolchain

Code Example

```
1 // Cross-platform C program with Cosmopolitan
2
3 #include <stdio.h> // Standard I/O header
4
5 /*
6  * Prints "Hello World" message to console
7  * Demonstrates portability across Linux/Windows/MacOS
8  * Can enable tracing with --strace or --ftrace
9  */
10
11 int main(int argc, char *argv[]) {
12
13     // Output message
14     printf("Hello World\n");
15
16     // Return success
17     return 0;
18
19 }
```

Run

Reset

Hello World